Riemann Roch Generalized

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Consider the Gauss-Bonnet theorem (this):

$$\int_M K dA = \int_{\partial M} k_g ds = 2\pi\chi(M)$$

where M is a 2-dimensional compact Riemann manifold, K the Gaussian curvature, k_g the geodesic curvature, and $\chi(M)$ the Euler characteristic. This can be thought of as a special case of The Riemann-Roch Theorem!

There is also a more direct topological generalization of the Gauss-Bonnet theorem known as the Chern theorem, or Chern–Gauss–Bonnet theorem, to $2n$-dimensional Riemann manifolds without boundary:

$$\chi(M) = \int_M e(\Omega)$$

where $\chi(M)$ is the Euler characteristic, Ω is the curvature form (or Levi-Civita connection), and $e(\Omega)$ is the Euler class:

$$e(\Omega) = \frac{1}{(2\pi)^n} \operatorname{Pf}(\Omega)$$

where Pf if the *Pfaffian*; namely as Ω can always be represented as a skew-symmetric matrix (when the dimension of M is even, hence the $2n$) it can always be written as the square of a polynomial in matrix entries, namely if A is the skew-symmetric matrix:

$$\operatorname{Pf}(A)^2 = \det(A)$$

for example:

$$A = \begin{pmatrix} 0 & a \\ -a & b \end{pmatrix} \qquad \operatorname{Pf}(A) = a$$

$$B = \begin{pmatrix} 0 & a & b & c \\ -a & 0 & d & e \\ -b & -d & 0 & f \\ -c & -e & -f & 0 \end{pmatrix} \qquad \operatorname{Pf}(B) = af - be + dc$$

Atiyah–Singer index theorem seems to generalize what I'm working on further.

Rapid fire concepts we know:

- We shall take varieties and schemes for granted
- We shall be working with algebraic curves, usually algebraic plane curves; hence curves given by $f(x, y)$ for coordinate functions x, y.
- Such curves are projective ([1]). If they are smooth, we naturally ask about their relation to one-dimension complex manifolds; these are called Riemann surfaces (as they are 2 real dimensional)

- Compact Riemann surfaces are classified up to homeomorphism by their geus, and $\chi(X) = 2 - 2g$ where χ is the Euler Characteristic.
- Via normalization, we have the following (This is in fact proven in [2]

Theorem 0.0.1: Characterizing Riemann Surfaces Using Normalization

Any comapct Riemann surcae $\tilde{C}$ can be obtain through the normalization of a certain plane algebraic curve C wit hat most ordinary double points, i.e. There exists a holomorphic mapping $\sigma : \tilde{C} \to \mathbb{P}^2(\mathbb{C})$ such that $\sigma(\tilde{C})$ is an algebraic curve possessing at most ordinary double points

- The geometry polynomial function (by FTC) makes them surjective and n-to-1 except at the ramification points.
- We shall take holomorphic and meromorphic differential forms for granted
- Key result about holomorphic differential forms:

Theorem 0.0.2: Stokes Theorem For Holomoprhic Differentials

Let C be a Riemann surface, $\Omega \subseteq C$ an open set, $\overline{\Omega}$ compact and $\partial\Omega = \gamma$ a piecewise smooth curve, and ‘ a holomorphic differential defined on an open set containing $\overline{\Omega}$. Then:

$$\int_{\partial\Omega} \omega = 0$$

Proof **:**
use local coordinates and apply Cauchy's theorem to get

$$\int_{\partial\Omega_i} \omega = 0$$

on the components, which then gluing gives the desired result

- Residue is winding number

1

Gallier Complex Analysis

this chapter compiles the important results from Gallier.

1.1 Riemann Roch in Modern Language

here

1.2 Generalized Riemann Roch: Vector Bundles

Let X be a complex irreducible algebraic variety (not necessarily smooth). Define:

1. X_{Zar}: X with the Zariski topology (abbreviated to Z-topology) and $\mathcal{O}_X$ the germs of algebraic functions
2. $X_{\mathbb{C}}$: X with the complex topology (abbreviated to $\mathbb{C}$-topology)and $\mathcal{O}_X$ the germs of algebraic functions
3. X^{an}: X with the complex topology and $\mathcal{O}_X$ the germs of holomorphic functions
4. X_{smooth}: X with the complex topology and $\mathcal{O}_X$ the germs of C^∞-functions

Then a vector bundle $p : E \to X$ can either be locally trivial in the $\mathbb{Z}$-topology or the $\mathbb{C}$-topology. As X is irreducible, it is connected, so the rank is a well-defined. If $E \to X$ has rank r, then:

1. (covering) there exists an open covering $\{U_\alpha \to X\}_\alpha$ along with isomorphisms:

$$\varphi_\alpha : p^{-1}(U_\alpha) \to U_\alpha \times \mathbb{C}^r$$

We shall sometimes denote $p^{-1}(U_\alpha)$ as $E|_{U_\alpha}$.

2. (local trivialization) For every α, the following diagram commutes:

$$\begin{tikzcd} p^{-1}(U_\alpha) \arrow[rr, "\sim"] \arrow[dr, "p"'] & & U_\alpha \times \mathbb{C}^r \arrow[dl, "\pi_1"] \\ & X & \end{tikzcd}$$

3. (gluing – cocycle condition)

$$\begin{tikzcd} p^{-1}(U_\alpha) \arrow[r, "\phi_\alpha"] \arrow[d] & U_\alpha \prod \mathbb{C}^r \arrow[d] \\ p^{-1}(U_\alpha \cap U_\beta) \arrow[r, "\phi_\alpha"] \arrow[d, equal] & (U_\alpha \cap U_\beta) \prod \mathbb{C}^r \arrow[d, "g_\alpha^\beta"] \\ p^{-1}(U_\beta \cap U_\alpha) \arrow[r, "\phi_\beta"] \arrow[d] & (U_\beta \cap U_\alpha) \prod \mathbb{C}^r \arrow[d] \\ p^{-1}(U_\beta) \arrow[r, "\phi_\beta"] & U_\beta \prod \mathbb{C}^r \end{tikzcd}$$

where by commutativity $g_\alpha^\beta = \varphi_\beta \circ \varphi_\alpha^{-1}|_{p^{-1}(U_\alpha \cap U_\beta)}$.

Note then that

$$g_\alpha^\beta|_{U_\alpha \cap U_\beta} = \mathrm{id} \qquad \text{and} \qquad g_\alpha^\beta|_{\mathbb{C}^r} \in \mathrm{GL}_r(\Gamma(U_\alpha \cap U_\beta), \mathcal{O}_X)$$

On a triple overlap, we get:

$$g_\beta^\gamma \circ g_\alpha^\beta = g_\alpha^\beta \qquad \text{and} \qquad (g_\beta^\alpha)^{-1} = g_\alpha^\beta$$

Hence, the collection $\{g_\alpha^\beta\}$ forms a 1-cocycle in $Z^1(\{U_\alpha \to X\}, \mathbb{GL})$ where $\mathbb{GL}$ or $\mathbb{GL}(X)$ is:

$$\Gamma(U, \mathbb{GL}_r(X)) = \mathrm{GL}_r(\Gamma(U, \mathcal{O}_X))$$

where the right hand side is the group of invertible linear maps on the free module $\Gamma(U, \mathcal{O}_X)^r \cong \Gamma(U, \mathcal{O}_X^r)$. When $r = 1$, we often denote $\mathbb{GL}_1(X)$ as $\mathbb{G}_m$ or $\mathcal{O}_X^*$.

Let's talk about choice. Say $\{\psi_\alpha\}_\alpha$ is another trivialization. Without loss of generality we may refine each cover so that $\{\varphi_\alpha\}_\alpha$ and $\{\psi_\alpha\}_\alpha$ use the same cover. Then there exists an isomorphism: $\sigma_\alpha : U_\alpha \times \mathbb{C}^r \to U_\alpha \times \mathbb{C}^r$ where the following diagram commutes:

$$\begin{tikzcd} & U_\alpha \times \mathbb{C}^r \arrow[dd, "\sigma_\alpha"] \\ p^{-1}(U_\alpha) \arrow[ur, "\varphi_\alpha"] \arrow[dr, "\psi_\alpha"'] & \\ & U_\alpha \times \mathbb{C}^r \end{tikzcd}$$

Then $\{\sigma_\alpha\}_\alpha$ form 0-cochains in $C^0(\{U_\alpha \to X\}, \mathbb{GL}_r)$. if $\{h_\alpha^\beta\}$ is the gluing information for the $\{\psi_\alpha\}$ then:

$$\phi_\beta = g_\alpha^\beta \circ \phi_\alpha \psi_\beta \qquad = h_\alpha^\beta \circ \psi_\alpha \psi_\alpha = \sigma_\alpha \circ \phi_\alpha.$$

Thus: $\sigma_\beta \circ \phi_\beta = \psi_\beta = h_\alpha^\beta \circ \sigma_\alpha \circ \phi_\alpha$, so:

$$\phi_\beta = (\sigma_\beta^{-1} \circ h_\alpha^\beta \circ \sigma_\alpha) \circ \phi_\alpha,$$

and so the transition between g_α^β and h_α^β is:

$$g_\alpha^\beta = \sigma_\beta^{-1} \circ h_\alpha^\beta \circ \sigma_\alpha$$

We thus get a conjugation equivalence relation $\sim$ on $Z^1(\{U_\alpha \to X\}, \mathbb{GL}_r)$ which trivializes co-cycles:

$$Z^1/\sim = H^1(\{U_\alpha \to X\}, \mathbb{GL}_r)$$

This is not a group as $\mathbb{GL}_r$ is not abelian, however it is a pointed set (the trivial bundle is singled out, pointed out). By passing through the limit over the covers by refinement, we can label the limit $\check{\mathrm{H}}^1(X, \mathbb{GL}_r)$ and we then have:

Theorem 1.2.1: Vector Bundle Classification Through Cohomology

Let X be an algebraic curve of type (I)-(IV). Then the set of isomorphisms classes of rank r of vector bundles is isomorphic to $\check{\mathrm{H}}^1(X, \mathbb{GL}_r)$

Proof **:**
the above work

This group is indeed the right choice, for if $1 \to G' \to G \to G'' \to 1$ is an exact sequence of of sheaves of groups (not necessarily commutative), then

$$\begin{tikzcd} 1 \arrow[r] & G'(X) \arrow[r] & G(X) \arrow[r] & G''(X) \arrow[dll, rounded corners, "\delta_0"] \\ & \check{H}^1(X, G') \arrow[r] & \check{H}^1(X, G) \arrow[r] & \check{H}^1(X, G'') \end{tikzcd}$$

is an exact sequence of pointed sets. To compute $\delta_0(\sigma)$ where $\sigma \in G''(X)$, proceed as follows: Cover X by suitable U_α and pick $s_\alpha \in G(U_\alpha)$ mapping to $\sigma|U_\alpha$ in $G''(U_\alpha)$. Set

$$\delta_0(\sigma) = s_\alpha s_\beta^{-1} \text{ on } U_\alpha \cap U_\beta/\sim.$$

Then $\delta_0(\sigma) \in \check{\mathrm{H}}^1(X, G')$.

Note that we took $\mathbb{GL}_r(X)$, but we could have taken the automorphisms of any object, namely if F is an object and $\mathrm{Aut}(F)$ is the automorphisms, we then call the X-torsor for F an "object E over X" locally (on X) of the form $U \times F$ and glued by paired (id, g) where $g \in \mathrm{Mor}(U \cap V, \mathrm{Aut}(F))$. Then we just classified all X-torsors of F.

(3)

$$0 \longrightarrow \mathbb{G}_m \longrightarrow \mathbb{G}_{r+1} \longrightarrow \mathbb{PGL}_r \longrightarrow 0 \text{ is exact.}$$

I'll finish later. the point is that we can look at iso of vect bundles over either Zariski or Complex topology where $\mathscr{F} = \mathscr{O}_X$ is alg functions and it is isomorphic; or holo functions over Zariski topology injects into over complex topology.

1.3 Terms for G-bundles

1. Principal G-bundle generalization of $X \times G \to G$. Let $P \to X$ be a whose fibers $p^{-1}(x)$ have a G-actions $G \times p^{-1}(x) \to p^{-1}(x)$ that is free and transitive. Given the appropriate topology, G is homeomorphic to $p^{-1}(x)$. The canonical example is the frame-bundle of a manifold M isomorphic to the appropriate subgroup of $\mathrm{GL}_r(\mathbb{R})$. One application of principal bundles is detecting when a manifold may have additional structures; commonly it would be that there exists a quotient M/G (solggeojk quotient per fiber) that has a complex, quaternion, spin, etc, structure.

2. Principal G-bundles also important for Čheck cohomology, as it has the transition information.

1.4 Hodge stuff

Let X be a complex analytic manifold of complex dimension n. As a real manifold ,it is a smooth manifold of dimension $2n$. Let TX be the tangent bundle and $T_{X,x}$ the fiber at x, i.e. the $\mathbb{C}$-vector space of (complex) dimension n. Let $z_1, ..., z_n$ be complex local coordinates at $x \in X$ on U; these are associated to $2n$ real coordinates $x_1, y_1, ..., x_n, y_n$ where $z_j = x_j + iy_j$. Given such $2n$ real coordinates, there is two natural coordinates that can be from them; z_j and $\overline{z}_j$ via:

$$z_j = x_j + iy_j \qquad \overline{z}_j = x_j - iy_j$$

Note that the z_j and $\overline{z}_j$ both form complex coordinates; the transition functions will be holomorphic as they are "consistent". Conversely, using the two sets of coordiantes (the complex and anti-complex) we can form:

$$x_j = \frac{1}{2}(z_j + \overline{z}_j) \qquad y_j = \frac{1}{2i}(z_j - \overline{z}_j)$$

If we treat TX_x as an $\mathbb{R}$-vector space, we can complexity to naturally combine the z_j and $\overline{z}_j$ coordinates. To see this, a $\mathbb{C}$-basis at $T_{X,x}$ is

$$\frac{\partial}{\partial z_1}, ..., \frac{\partial}{\partial z_n}$$

And $\mathbb{R}$-basis at $T_{X,x}$ is

$$\frac{\partial}{\partial x_1}, \frac{\partial}{\partial y_1}, ..., \frac{\partial}{\partial x_n}, \frac{\partial}{\partial y_n}$$

which is real $2n$-dimensional (and each pair $\partial/\partial x_j$, $\partial/\partial y_j$ satisfy the Cauchy-Riemann equation). We can create complex dimensional spaces by tensoring:

$$T_{X,x_{\mathbb{C}}} = T_{X,x} \otimes_{\mathbb{R}} \mathbb{C}$$

Then $T_{X,x_{\mathbb{C}}}$ has a basis

$$\frac{\partial}{\partial z_j}, \frac{\partial}{\partial \bar{z}_j} \qquad 1 \leq j \leq n$$

Then for any $f \in C^\infty(U)$:

$$\frac{\partial f}{\partial z_j} = \frac{\partial f}{\partial x_j} - i\frac{\partial f}{\partial y_j} \qquad \text{and} \qquad \frac{\partial f}{\partial \bar{z}_j} = \frac{\partial f}{\partial x_j} + i\frac{\partial f}{\partial y_j}$$

This is what is going on in coordinates, but we can also view the theory more abstractly. Let $(e_1, ..., e_n)$ be a $\mathbb{C}$-basis for V. Then so is $(ie_1, ..., ie_n)$. We can have an $\mathbb{R}$-basis for V via $(f_1, ..., f_n, g_1, ..., g_n)$ and write $e_j = f_j + ig_j$. Then the map $(e_1, ..., e_n) \mapsto (ie_1, ..., ie_n)$ is represented in the $\mathbb{R}$-basis as :

$$((f_1, g_1), ..., (f_n, g_n)) \xrightarrow{J} ((-g_1, f_1), ..., (-g_n, f_n))$$

We can view J as an $\mathbb{R}$-endomorphism of V with the property that $J^2 = -\mathrm{id}$. Does this in fact characterize complex multiplication? If we have an $\mathbb{R}$-space of dimension $2n$ and an $\mathbb{R}$-endomorphism J of V such that $J^2 = -\mathrm{id}$, then we can give V a complex structure (i.e. a $\mathbb{C}$) action via:

$$(a + bi) \cdot v = av + bJ(v)$$

For example, on $\mathbb{R}^2$, there is $J(a_1, a_2) = (-a_2, a_1)$ and $J(a_1, a_2) = (a_2, -a_1)$. It is an exercise in algebra to see that on a real $2n$-dimensional vector space V, the $\mathbb{C}$-vector space structures on V corresponding bijective to the homogeneous to $\mathrm{GL}_{2n}(\mathbb{R})/\mathrm{GL}_n(\mathbb{C})$ via:

$$[A] \mapsto AJA^{-1}$$

Definition 1.4.1: Almost Complex Structure

Let X be a real smooth manifold. Then it has an *almost complex structure* if they exists a bundle morphism $J : TX \to TX$ such that $J^2 = -\mathrm{id}$

Proposition 1.4.2: Complex Analytic, then Almost Complex

Let $(X, \mathcal{O}_X)$ be a complex analytic manifold. Then it is almost complex

Proof **:**
exercise (it is just pushing definitions around)

Now, consider a complex n-dimensional complex bundle TX; let $z_1, ..., z_n$ be its coordinate. Then as mentioned earlier, we can complexity to get a basis:

$$\frac{\partial}{\partial z_1}, ..., \frac{\partial}{\partial z_n}, \frac{\partial}{\partial \bar{z}_1}, ..., \frac{\partial}{\partial \bar{z}_n}$$

and so, in this basis, if we write $f = (w_1, \dots, w_n)$, where $w_\alpha = u_\alpha + iv_\alpha$, we get

$$J_{\mathbb{R}}(f) = \begin{pmatrix} \left(\frac{\partial w_\alpha}{\partial z_j}\right) & \left(\frac{\partial w_\alpha}{\partial \bar{z}_j}\right) \\ \left(\frac{\partial \overline{w}_\alpha}{\partial z_j}\right) & \left(\frac{\partial \overline{w}_\alpha}{\partial \bar{z}_j}\right) \end{pmatrix}$$

And if f is holomorphic:

$$\frac{\partial w_\alpha}{\partial \bar{z}_j} = \frac{\partial \bar{w}_\alpha}{\partial z_j} = 0,$$

giving:

$$J_{\mathbb{R}}(f) = \begin{pmatrix} \left(\frac{\partial w_\alpha}{\partial z_j}\right) & 0 \\ 0 & \left(\frac{\partial \bar{w}_\alpha}{\partial \bar{z}_j}\right) \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & \bar{A} \end{pmatrix}.$$

Now, write:

$$J(f) = \left(\frac{\partial w_\alpha}{\partial z_j}\right)$$

and call it the *holomorphic Jacobian*. Then:

1. $J_{\mathbb{R}}(f) = \begin{pmatrix} J(f) & 0 \\ 0 & \overline{J(f)} \end{pmatrix}$, so, $\mathbb{R}$-rank $J_{\mathbb{R}}(f) = 2\mathbb{C}$-rank $J(f)$.
2. We have $\det(J_{\mathbb{R}}(f)) = |\det(J(f))|^2 \geq 0$, and $\det(J_{\mathbb{R}}(f)) > 0$ if f is a holomorphic isomorphism (in this case, $m = n$ the common dimension of X, Y).

Then we see holomorphic maps preserve the orientation of a complex manifold and each complex manifold posses an orientation.

There is a lot more about Hodge stuff, but I don't feel like this is super helpful right now

I will put here what I think is the most important results for now

2

Key Results from Topological Methods from Algebraic Geometry

This will compile the results from Hirzebruch's textbook that may be important.

2.1 Some Hodge Theory

Let X be a smooth complex projective manifold ost hat X is a submanifold of $\mathbb{C}\mathrm{P}^n$ and by Chow's theorem algebraic. The Riemann metric on $\mathbb{C}\mathrm{P}^n$ induces a metric on X. This metric is "strongly compatible" with the complex structure, in particular it is a Kähler manifold.

Any 1-form with complex coefficients can be uniquely with respect written as a sum:

$$\sum_{j=1}^{n}(f_j dz^j = g_j d\overline{z}^j)$$

Then we know that for every $r \in \mathbb{N}$, every smooth r-form on X (with $\mathbb{C}$-coefficients) can be written as the sum of (p,q)-forms where $p+q=r$. In particular, let:

$$\Omega^{p,q} = \underbrace{\Omega^{1,0} \wedge \cdots \wedge \Omega^{1,0}}_{p \text{ times}} \wedge \underbrace{\Omega^{0,1} \wedge \cdots \wedge \Omega^{0,1}}_{q \text{ times}}$$

where $\Omega^{1,0}$ is the $\mathbb{C}$-space generated locally by dz and $\Omega^{0,1}$ generated locally by $d\overline{z}$. The spaces $\Omega^{1,0}$ and $\Omega^{0,1}$ are bot stable under holomorphic coordinate changes (use Cauchy-Riemann).

Now, let E^k be the space of all complex differential forms of total degree k. Then

$$E^k = \Omega^{k,0} \oplus \Omega^{k-1,1} \oplus \cdots \oplus \Omega^{1,k-1} \oplus \Omega^{0,k} = \bigoplus_{p+q=k} \Omega^{p,q}$$

When the manifold is nice enough, ex. Kähler, $\Omega^k = E^k$ (where Ω^k is the collection of holomorphic forms, "decomposition" like we did above, that is the *Hodge decomposition*). Note that the direct sum decomposition is stable under holomorphic coordinate changes, hence it determines a vector bundle decomposition. Note we get a natural projection for each direct summand:

$$\pi^{p,q} : E^K \to \Omega^{p,q}$$

Now, define $d : \Omega^r \to \Omega^{r+1}$ in the natural way where:

$$d(\Omega^{p,q}) \subseteq \bigoplus_{r+s=p+q+1} \Omega^{r,s}$$

When we were just working with smooth manifolds, this gave us "enough" cohomological information. We now have more structure and this is not granular enough. We define the *Dolbeault operators*:

In coordinates, let:

$$\alpha = \sum_{|I|=p,|J|=q} f_{IJ} dz^I \wedge d\bar{z}^J \in \Omega^{p,q}$$

where I and J are multi-indices. Then:

$$\partial\alpha = \sum_{|I|,|J|} \sum_{\ell} \frac{\partial f_{IJ}}{\partial z^\ell} dz^\ell \wedge dz^I \wedge d\bar{z}^J$$

and

$$\bar{\partial}\alpha = \sum_{|I|,|J|} \sum_{\ell} \frac{\partial f_{IJ}}{\partial \bar{z}^\ell} d\bar{z}^\ell \wedge dz^I \wedge d\bar{z}^J.$$

Furthermore:

$$d = \partial + \bar{\partial}$$

$$\partial^2 = \bar{\partial}^2 = \partial\bar{\partial} + \bar{\partial}\partial = 0.$$

Thus, we see there is some interesting cohomology that can occur; this defines *Dolbeault cohomology.* The symbol ∂ suggests that they are dual to d, and indeed the Poincaré lemma generalizes to complex manifolds.

Let us define the Dolbeault cohomology. Notice that $\overline{\partial} f = 0$ if f is holomorphic, therefore it is in some sense the "natural" choice; the other choice keeping track of the "anti-holomorphic" information. Thus, define:

$$H^{p,q}(M) = \frac{\ker(\overline{\partial} : \Omega^{p,q} \to \Omega^{p,q+1})}{\operatorname{im}(\overline{\partial} : \Omega^{p,q-1} \to \Omega^{p,q})}$$

If we wanted to be precise, we could say $H^{p,q}_{\overline{\partial}}(M)$. As

$$H^{p,q}_{\overline{\partial}}(M) = \overline{H^{p,q}_{\partial}(M)}$$

We see that the difference of choice is a difference in conjugation, and hence we stick to $\overline{\partial}$ for the reasons mentioned before.

For our purposes, we will naturally want to generalize this to the Dolbeault cohomology of vector bundles. let E be a holomorphic vector bundle on a complex manifold X. Then take the map:

$$\overline{\partial}_E : \Gamma(E) \to \Omega^{0,1}(E)$$

taking section of E to $(0,1)$-forms with values in E. This will satisfy the Leibniz rule with respect to $\overline{\partial}$, and is thus a $(0,1)$-connecting on E. Then we can do the same trick from Riemann geometry and extended it to the exterior covariant derivative, namely;

$$\overline{\partial}_E : \Omega^{p,q}(E) \to \Omega^{p,q+1}(E)$$

which acts on on a section $\alpha \otimes s \in \Omega^{p,q}(E)$ by:

$$\overline{\partial}_E(\alpha \otimes s) = (\overline{\partial}\alpha) \otimes s + (-1)^{p+q}\alpha \wedge \overline{\partial}_E s$$

This naturally extends linearly to any section of $\Omega^{p,q}(E)$ and $\overline{\partial}_E^2 = 0$. Thus, the Dolbeault cohomology with coefficient in E is:

$$H^{p,q}(X,(E,\bar{\partial}_E)) = \frac{\ker(\bar{\partial}_E : \Omega^{p,q}(E) \to \Omega^{p,q+1}(E))}{\operatorname{im}(\bar{\partial}_E : \Omega^{p,q-1}(E) \to \Omega^{p,q}(E))}$$

It is standard to check that the Dolbeault cohomology groups do not depend on the choice of Dolbeault operator $\overline{\partial}_E$ compatible with the holomorphic structure of E, and so it is simply denoted by:

$$H^{p,q}(X,E)$$

There is a ton of fascinating theory here which I will leave for when I explore hodge theory. Let Ω^p be the sheaf of holomorphic p-forms on M. Then:

$$H^{p,q}(M) \cong H^q(M,\Omega^p)$$

and if we are working over a vector bundle:

$$H^{p,q}(M,E) \cong H^q(M,\Omega^p \otimes E)$$

This is *Dolbeault's Theorem*. This also means that:

$$H^{0,q}(M,E) \cong H^q(M,\Omega^0 \otimes E) = H^q(M,E)$$

For future reference, if E is a complex vector bundle, we shall denote its conjugate by $\overline{E}$ (defined in the natural way) which is anti-complex and anti-isomorphic to E. We can also define the dual bundle E^* in the natural way. Note that per-fiber, the dual can be given by Hermitian of the inverse of the matrix.

Hirzebruch introduces the following notation:

$$A^{p,q}(W) = \Gamma(V,\Omega^{p,q}(W))$$

and:

$$A^{p,q} = A^{p,q}(\mathbb{1})$$

where $\mathbb{1}$ is the trivial bundle representing holomorphic functions. Let M now be compact. We shall define the Hodge operator. If W is a complex vector bundle, we can define W^* and a (perfect) pairing:

$$A^{p,q}(W) \times A^{r,s}(W^*) \to A^{p+r,q+s}(\mathbb{1}) \qquad (\alpha,\beta) \mapsto \alpha \wedge \beta$$

If $W = \mathbb{1}$, this is the usual wedge (or exterior) product. The product is anti-commutative in the sense that:

$$\alpha \wedge \beta = (-1)^{(p+r)(q+s)} \beta \wedge \alpha$$

Choosing $r = n - p$, $s = n - q$ we get the top form $\alpha \wedge \beta \in A^{n,n}(\mathbb{1})$. We can thus define:

$$\langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta$$

By Stokes theorem, if $\alpha = \overline{\partial}\gamma$ for $\gamma \in A^{p,q-1}(W)$ and $\overline{\partial}\beta = 0$, (or if $\overline{\partial}\alpha = 0$ and $\beta = \overline{\partial}\gamma$) then:

$$\langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta = \int_M \overline{\partial}(\gamma \wedge \beta) = \int_M d\,(\gamma \wedge \beta) = 0$$

Thus, we get a well-defined pairing:

$$H^{p,q}(M, E) \times H^{n-q,n-q}(M, E^*) \to \mathbb{C} \qquad (\alpha, \beta) \mapsto \langle \alpha, \beta \rangle = \int_M \alpha \wedge \beta$$

Now, by some linear algebra and differential geometry we see that $\overline{TM} \cong T^*M$. Then we get:

$$\begin{aligned}
\bigwedge^{p} TM \otimes \bigwedge^{q} TM &\cong \bigwedge^{p} TM \otimes \bigwedge^{n} T^*M \otimes \bigwedge^{n} TM \otimes \bigwedge^{q} T^*M \\
&\cong \bigwedge^{n-p} T^*M \otimes \bigwedge^{n-q} TM \\
&\cong \bigwedge^{n-q} TM \otimes \bigwedge^{n-p} \overline{TM}
\end{aligned}$$

There is thus a natural duality operator:

$$* : \bigwedge^{p} TM \otimes \bigwedge^{q} TM \to \bigwedge^{n-q} TM \otimes \bigwedge^{n-p} \overline{TM}$$

and $* \circ *$ becomes multiplication by $(-1)^{p+q}$.

Now, by vector-bundle theory we know we can without loss of generality reduce W to a unitary group bundle. There is then the natural hermitian anti-isomorphism:

$$\psi : W \to W^* \qquad \psi^{-1} : W^* \to W$$

If κ is the anti-isomorphism from $\bigwedge^r TM \otimes \overline{\bigwedge^s TM}$ to itself via conjugation, Then define:

$$\star = \psi \otimes (\kappa \circ *) \qquad \widetilde{\star} = \psi^{-1} \otimes (\kappa \circ *)$$

This gives:

$$\begin{aligned}
\star &: W \otimes \bigwedge^{p} TM \otimes \overline{\bigwedge^{q} TM} \to W^* \otimes \bigwedge^{n-p} TM \otimes \overline{\bigwedge^{n-q} TM} \\
\widetilde{\star} &: W^* \otimes \bigwedge^{r} TM \otimes \overline{\bigwedge^{s} TM} \to W \otimes \bigwedge^{n-r} TM \otimes \overline{\bigwedge^{n-s} TM}.
\end{aligned}$$

If $r = n - p$, $s = n - q$, then $\widetilde{\star} \circ \star$ is multiplication by $(-1)^{p+q}$. Now, these produce natural anti-isomorphisms of:

$$\begin{aligned}\star &: \Omega^{p,q}(W) \to \Omega^{n-p,n-q}(W^*)\\ \widetilde{\star} &: \Omega^{r,s}(W*) \to \Omega^{n-r,n-s}(W)\end{aligned}$$

Using these, we can define the "co-" of $\overline{\partial}$:

$$\delta : \Omega^{p,q}(W) \to \omega^{p,q-1}(W) \qquad \delta = -\widetilde{\star} \circ \overline{\partial} \circ \star$$

With this, if $\alpha, \beta \in A^{p,q}(W)$ then we can define:

$$(\alpha, \beta) = \langle \alpha, \star\beta \rangle = \int_M \alpha \wedge \star\beta$$

It is easy to see that $(\alpha, \alpha) \geq 0$ and $(\alpha, \alpha) = 0$ if and only if $\alpha = 0$. We also get the natural adjoint:

$$(\alpha, \delta\beta) = \langle \overline{\partial}\alpha, \beta \rangle$$

Indeed given:

$$\langle \alpha, \delta\beta \rangle = -\int_M \alpha \wedge \star\widetilde{\star}\overline{\partial}\beta = (-1)^{p+q+1} \int_M \alpha \wedge \overline{\partial} \star \beta$$

Then:

$$\begin{aligned}(\overline{\partial}\alpha, \beta) - (\alpha, \delta\beta) &= \int_M (\overline{\partial}\alpha \wedge \star\beta + (-1)^{p+q}\alpha \wedge \overline{\partial} \star \beta)\\ &= \int_M \overline{\partial}(\alpha \wedge \star\beta)\\ &= \int_M d(\alpha \wedge \star\beta)\\ &= 0 && \text{Stokes Thm.}\end{aligned}$$

We now reach one of the central results in Hodge theory. Define the *Hodge Laplace* (or Laplace-Bertram operator):

$$\Delta = \delta\overline{\partial} + \overline{\partial}\delta$$

This defines an operator:

$$\Delta : A^{p,q}(W) \to A^{p,q}(W)$$

Let

$$\ker \Delta = B^{p,q}(W)$$

which could be interpreted as all "complex harmonic forms", namely forms that satisfy:

$$\delta(\alpha) = \overline{\partial}(\alpha) = 0$$

The use of the elliptic operator is that it gives way to the following decomposition:

$$A^{p,q}(W) = \overline{\partial}A^{p,q-1}(W) \oplus \delta A^{p,q+1}(W) \oplus B^{p,q}(V, W)$$

Notice that we naturally get:

$$Z^{p,q}(W) = \overline{\partial} A^{p,q-1}(W) \oplus B^{p,q}(V,W)$$

Thus, when taking the Dolbeault cohomology group:

$$H^{p,q}(V,W) \cong \frac{Z^{p,q}(W)}{\overline{\partial} A^{p,q}(W)} \cong B^{p,q}(V,W)$$

Hence, the forms are naturally harmonic! This result is called the *Hodge Theorem*. Furthermore, as this is an elliptic operator, we get by elliptic operator theory that each of these spaces is finite dimensional. We thus also get:

$$H^r(M,\mathbb{C}) \cong \bigoplus_{p+q=r} B^{p,q}$$

and the Betti number is the sum of Hodge numbers:

$$b_r(V) = \sum_{p+q=r} h^{p,q}(M)$$

For future reference, an element of $H^{p+q}(M,\mathbb{Z})$ or $H^{p+q}(M,\mathbb{R})$ is said to be of type (p,q) if it is of type (p,q) when regarded as an element of $H^{p+q}(M,\mathbb{C})$.

All of this is summarized in the following two theorems:

Theorem 2.1.1: Kodaria's Finite Dimension Theorem

Then $H^{p,q}(M,E)$ is finite dimensional isomorphic to the vector space of complex harmonic forms of type (p,q) with coefficients in E. Furthermore:

$$H^p(M,E) = H^{0,p}(M,E)$$

and is zero for $p,q > n$.

***Proof* :**
see a textbook in Hodge theory

Theorem 2.1.2: Serre-Hodge Duality Theorem

Let E be a complex vector bundle over the compact manifold M. Then the bilinear form $\langle -,- \rangle$ is a dual pairing of $H^{p,q}(M,E)$ with $H^{n-p,n-q}(M,E^*)$. If $K = \bigwedge^n TM$ is the canonical line bundle, Then $H^p(M,E)$ and $H^{n-q}(M, K \otimes E^*)$ are dual vector spaces.

***Proof* :**
see a textbook in Hodge theory

Definition 2.1.3: Hodge Number

$$h^{p,q}(M,E) = \dim H^{p,q}(M,E)$$

If $E = \mathbb{1}$ then:

$$h^{p,q}(M) = \dim H^{p,q}(M,\mathbb{1})$$

Note that in general $h^{p,q}(M) \neq h^{q,p}(M)$; if M is a Kähler manifold this will be the case.

2.1.1 Defining Kähler manifold

Let M_n be a compact complex Riemann manifold. A hermitian metric on M is defined as:

$$ds^2 = 2\sum g_{\alpha\beta}(z,\overline{z})(dz^\alpha \cdot d\overline{z}^\beta)$$

where $\overline{g_{\alpha\beta}} = g_{\beta\alpha}$. To a Hermitian metric ds^2, we get a natural exterior differential form:

$$\omega = i\sum g_{\alpha\beta}(z,\overline{z})dz^\alpha \wedge d\overline{z}^\beta$$

This form can be re-written as a real differential form by taking real coordinates x^α, $1 \leq \alpha \leq 2n$ and

$$z^\alpha = x^{2\alpha-1} + ix^{2\alpha}$$

Definition 2.1.4: Kähler Metric

Let ds^2 be a Hermitian metric on the compact complex Riemann manifold M and let:

$$\omega = i\sum g_{\alpha\beta}(z,\overline{z})dz^\alpha \wedge d\overline{z}^\beta$$

Then if $d\omega = 0$, ds^2 is called a *Kähler metric.*

If ds^2 is a Kähler metric, ω represents an element of $H^2(M,\mathbb{R})$ (using the De-Rham isomorphism); this is called the *fundamental class of the Kähler metric.*

Definition 2.1.5: Kähler Manifold

Let M be a compact complex Riemann manifold with Riemann metric ds^2. Then given a choice of ω that induces ds^2 to be Kähler metric, (M,ω) is called a *Kähler manifold.*

On a Kähler manifold, Δ commutes with the conjugation $\alpha \mapsto \overline{\alpha}$ between $B^{q,p}$ and $B^{p,q}$ which induces an isomorphism between these groups, and so:

$$h^{p,q}(M) = h^{q,p}(M)$$

Note that these numbers do not depend on the choice of Kähler metric ($d\alpha = 0$ for a harmonic (p,q)-form regardless of metric) and hence it is enough for a Kähler metric to exist.

2.1.2 Euler Characteristic and Generalization

Let M be a smooth compact manifold, E a complex vector bundle (we shall introduce the Kähler assumption later). Now, as we have a double complex, we can define more than one Euler characteristic. First, recall there is the natural:

$$\chi(M,E)=\sum_{i=0}^{\infty}\dim H^i(M,E)=\sum_{i=0}^{n}\dim H^i(M,E)$$

What we can do now is chose a grading so that we can define:

$$\chi^p(M,E)=\chi\left(V,E\otimes\bigwedge^p TM\right)=\sum_{q=0}^{n}(-1)^q h^{p,q}(M,E)$$

Immediately we get:

$$\chi^0(M,E)=\chi(M,E)\qquad\text{and}\qquad\chi^p(M,E)=0\text{ when }p<0\text{ or }p>n$$

and if $E=\mathbb{1}$ then:

$$\chi^p(M,\mathbb{1})=\chi^p(M)=\sum_{q=0}^{n}(-1)^q h^{p,q}(M)$$

We can collect these characteristic in a generating function:

Definition 2.1.6: χ_y-characteristic

$$\chi_y(M,E)=\sum_{p=0}^{n}\chi^p(M,E)y^p\qquad\text{(sim.)}\qquad\chi_y(M)=\sum_{p=0}^{n}\chi^p(M)y^p$$

The term $\chi_y(M,E)$ is the χ_y-characteristic of E and $\chi_y(M)$ is called the χ_y-genus of M

By construction:

$$\chi_0(M,E)=\chi^0(M,E)=\chi(M,E)\qquad\text{(sim.)}\qquad\chi_0(M)=\chi^0(M)=\chi(M)$$

As the Euler-Hodge characteristic was called a genus, we shall call:

$$\chi(M)=\sum_{q=0}^{n}(-1)^q h^{0,q}(M)$$

the *arithmetic genus* of M. By Serre Duality:

$$\begin{aligned}\chi^p(M,E)&=(-1)^n\chi^{n-p}(M,E^*)\\ \chi(M,E)&=(-1)^n\chi(M,K\otimes E^*)\end{aligned}$$

Let us now assume M is Kähler. Let $g_q=h^{q,0}$. Then by the symmetry, the arithmetic genus $\chi(M)$ of a compact Kähler manifold is equal to:

$$\chi(M)=\sum_{i=0}^{n}(-1)^i g_i$$

Next, let us understand $\chi_y(M)$ for specific values of y. Let us start with $y = -1$. Then:

$$\chi_{-1}(M) = \sum_{p=0}^{n} (-1)^p \chi^p(M) = \sum_{p,q} (-1)^{p+q} h^{p,q}(M)$$

is equal to the Euler-Poincaré characteristic.

If V_n and V'_m are Kähler manifolds then

$$h^{p,q}(V_n \times V'_m) = \sum_{\substack{r+u=p \\ s+v=q}} h^{r,s}(V_n) h^{u,v}(V'_m).$$

Let y, z be indeterminates and associate to each Kähler manifold V the polynomial $\prod_{y,z}(V) = \sum_{p,q} h^{p,q} y^p z^q$. Then the above is equivalent to:

$$\prod_{y,z}(V_n \times V'_m) = \prod_{y,z}(V_n) \cdot \prod_{y,z}(V'_m).$$

Let $z = -1$ in. Then $\prod_{y,-1} = \chi_y$ and

$$\chi_y(V_n \times V'_m) = \chi_y(V_n) \cdot \chi_y(V'_m),$$

which is a property common to χ_y and T_y.

Now, with $y = 1$, then:

Theorem 2.1.7: Euler Characteristic With $y = 1$

$$\chi_1(M) = \sum_{p=0}^{n} \chi^p(M) = \sum_{p,q} (-1)^q h^{p,q}(M)$$

is equal to the *index* $\tau(M)$ (defined in [3, p. 8.2]).

Proof **:**
long proof at [3, p.125, Thm 15.8.2]

This will be a key step in the proof of RR. Note that there is a more direct proof for the above that works for arbitrary compact complex manifolds M (Hirzebruch sketches a proof in [3, appendix 25.4]).

Let us take a moment to understand the first Chern class on a Kähler manifold, considering it will be key for the RR on line bundles. Take the usual exact sequence:

$$0 \to \mathbb{Z} \to \mathcal{O}_M \to \mathcal{O}_M^* \to 0$$

where $\mathcal{O}_M$ is the sheaf of holomorphic functions, and $\mathcal{O}_M$ is the sheaf of never=zero holomorphic functions (invertible holomorphic functions). Then we get an exact cohomology sequence:

$$H^1(M, \mathcal{O}_M^*) \xrightarrow{\delta_*^1} H^2(M, \mathbb{Z}) \to H^2(M, \mathcal{O}_M^*)$$

Then on a Kähler manifold we have $H^2(M,\mathcal{O}_X^*) = H^2(M,\mathbb{1}) \cong B^{0,2}(M)$. Thus, an element $a \in H^2(M,\mathbb{Z})$ is mapped to a nonzero element of $H^2(M,\mathcal{O}_M)$ if and only if a is of type $(1,1)$. Then from characteristic theory for vector bundles we know that if $\xi \in H^1(M,\mathcal{O}_M^*)$ is a complex $\mathbb{C}^*$-bundle, then:

$$\delta_*^1(\xi) = c_1(\xi)$$

If F is a complex line bundle over M and ξ is the assocaited $\mathbb{C}^*$-bundle, then $c_1(\xi)$ is the *cohomology class* of F. Then we get:

Theorem 2.1.8: characterising Line bundles on Kähler Manifolds

Let M be a compact Kähler manifold. Then an element $a \in H^2(M,\mathbb{Z})$ is the cohomology class of a complex lien bundle over M if and only if it is of type $(1,1)$.

***Proof* :**
Hirzebruch references ppl at [3, p.127, Thm 15.9.1], and says it is provable for non-Kähler manifolds and was done by Dolbeault.

Let us finish with the special case M a Kähler manifolds and $h^{p,q} = 0$ when $p \neq q$. Then $\chi_y(M)$ is (essentially) equal to the poincaré polyomial:

$$P(t;M) = \sum b_r t^r$$

where b_r is the rth Betti number of M. Indeed, we get:

$$\chi^p(M) = \sum_q (-1)^1 h^{p,q} = (-1)^p h^{p,p} = (-1) b_{2p}$$

therefore:

$$\chi_{-t^2}(M) = P(t;V) = \sum_r b_r t^r$$

Why do we care about this special case? Because complex projectives varieties $\mathbb{C}\mathrm{P}^n$ and flag manifolds $F(n)$ both are Kähler manifolds with this property. The $\mathbb{C}\mathrm{P}^n$ comes from basic Dolbeault cohomology. For the flag manifold $F(n)$, recall that $H^*(F(n),\mathbb{Z})$ by the splitting principle is generated by element $\gamma_i \in H^2(F(n);\mathbb{Z})$. Then by characteristic theory there are complex $\mathbb{C}^*$-bundles ξ_i over $Fn)$ where $c_1(\xi_i) = \gamma_i$. The "only if" comes from theorem 2.1.8 where we have that each γ_i is of type $(1,1)$, and therefore the cohomology classes of $F(n)$ are of type (p,p).

For future reference, notice that χ_y and T_y ([3, p. 14.4]) agree, as they are both essentially equal to the Poincaré polynomial.

2.1.3 Properties of χ_y-characteristic

Part of the proof will requires some splitting properties that we can induct on. To that end, we shall develop some more properties of χ_y. Let :

$$0 \to E' \to E \to E'' \to 0$$

be a short exact sequence of complex vector bundles. This naturally induces a short exact sequence of sheaves:

$$0 \to \Omega(E') \to \Omega(E) \to \Omega(E'') \to 0$$

which is left as an exercise in sheaf theory. What is important is that we get that we get that each new Euler-characteristic is additive on exact sequences:

Theorem 2.1.9: Generalized Euler Characteristic Additive

Let

$$0 \to E' \to E \to E'' \to 0$$

be a short exact sequence of complex vector bundles over M. Then:

$$\begin{aligned} \chi(M, E) &= \chi(M, E') + \chi(M, E'') \\ \chi^p(M, E) &= \chi^p(M, E') + \chi^p(M, E'') \\ \chi_y(M, E) &= \chi_y(M, E') + \chi_y(M, E'') \end{aligned}$$

Proof **:**
exercise.

Theorem 2.1.10: Splitting Principle and Euler Characteristic

Let E be a complex vector bundle over a compact complex manifold M, and suppose the structure group of E can be complex analytically reduced to the triangular group $\Delta(q, \mathbb{C})$. Let $A_1, A_2, ..., A_q$ be the corresponding diagonal line bundles. Let E' be another complex vector bundle over M. Then:

$$\chi(M, E' \otimes E) = \sum_i^q \chi(M, E' \otimes A_i)$$

Proof **:**
Use induction and the above theorem

Let us now get some important ways Cartier divisors come into the picture. As usual let E be a complex vector bundle over a complex manifold M, and let D be a nonsingular divisor of M. Then it is associate to a line bundle $\mathcal{O}_M(D) = \mathcal{L}(D)$. We can now tensor $E \otimes \mathcal{L}(D)$, and consider $(E \otimes \mathcal{L}(D))_D$ to be the restriction to D of the vector bundle $E \otimes \mathcal{L}(D)$. We can take the sheaf $\Omega((E \otimes \mathcal{L}(D))_D)$ over D. Extending by zero this is a sheaf on M. By the corresponding of vector bundles and sheaf, without loss of generality we may denote this sheaf by E_D.

Theorem 2.1.11: Extending via Divisors

let M be a complex manifold , D a nonsingular Cartier divisor of M, E a complex vector bundle over M. Then there is a short exact sequence:

$$0 \to \Omega(E) \to \Omega(E \otimes \mathcal{L}(D)) \to E_D \to 0$$

***Proof* :**
This is a standard proof, this can be found in pure-sheaf form in Hartshorne [5, chapter 2?] or EYTNKA alg geo [1].

Let M now be compact and D still a nonsingular divisor. It now is always a compact manifold. When it is not confusing, we shall denote E_D to be the complex vector bundle over D. We shall also write:

$$\chi(D,E) := \chi(D,E_D),\ \chi^p(D,E) := \chi^p(D,E_D),\ \chi_y(D,E) := \chi_y(D,E_D)$$

Theorem 2.1.12: Splitting Euler Characteristic Using Divisor

Let M be a compact complex manifold, D a nonsingular divisor, E a complex vector bundle over M. Then:

$$\chi(M,E) = \chi(M, E\otimes \mathcal{L}(D)^{-1}) + \chi(D,E)$$

In particular, if $E = \mathbb{1}$, then:

$$\chi(M) = \chi(M, \mathcal{L}(D)^{-1}) + \chi(D)$$

***Proof* :**
combine above results

Let us now apply these results to get some important identities. First, recall from sheaf theory that we get the following short exact sequence:

$$0 \to T(D) \to T(M)|_D \to \mathcal{L}(D)_D \to 0$$

We then get a natural exact sequence:

$$0 \to \bigwedge^p (T(D)) \to \bigwedge^p (T(M)|_D) \to \bigwedge^{p-1} (T(D)) \otimes \mathcal{L}(D)_D \to 0$$

Dualizing we get:

$$0 \to \bigwedge^{p-1} (T^*(D)) \otimes \mathcal{L}(D)_D^{-1} \to \bigwedge^p (T^*(M)|_D) \to \bigwedge^p (T^*(D)) \to 0$$

Now if E is a complex vector bundle over M, we can tensor each term by E_D. Then we get:

$$\chi\left(D, E\otimes \bigwedge(T^*(M))\right) = \chi^{p-1}(D, E\otimes\mathcal{L}(D)^{-1}) + \chi^p(D,E)$$

Replacing E with $E\otimes \bigwedge^p(T^*(M))$, manipulating we get:

$$\chi^p(M,E) = \chi^p(M, E\otimes\mathcal{L}(D)^{-1}) + \chi^p(D,E) + \chi^{p-1}(D, E\otimes \mathcal{L}(D)^{-1}) \tag{2.1}$$

This is true for $p > 0$, and for $p = 0$ interpret the last term on the right hand side as being 0. Furthermore, $\chi^p(D,E) = 0$ for $p = n = \dim M$ and if $p > n$ then all four terms are 0. Letting y be a indeterminate, the above equation holds with each term multiplied by y^p. Summing over $p \geq 0$ gives:

$$\chi_y(M, E) = \chi_y(M, E \otimes \mathcal{L}(D)^{-1}) + \chi_y(D, E) + y\chi_y(D, E \otimes \mathcal{L}(D)^{-1})$$

Now, by repeating the application of equation (2.1), we can write $\chi^p(D, E)$ as a linear combination of integers of the form $\chi^q(M, A)$ where A is will be an appropriate? Hirzebruch just says certain?? complex vector bundle over M. For example:

$$\chi^0(D, E) = \chi^0(M, E) - \chi^0(M, E \otimes \mathcal{L}(D)^{-1}) \tag{2.2}$$

To calculate the last term, we use Serre duality and use:

$$\chi_1(M, E) = \chi_1(M, E \otimes \mathcal{L}(D)^{-1}) + \chi_1(D, E) + \chi_1(D, E \otimes \mathcal{L}(D)^{-1})$$

and replace E with $E \otimes \mathcal{L}(D)^{-1}$ in equation (2.2) to get:

$$\chi^1(D, E) = \chi^1(M, E) - \chi^1(M, E \otimes \mathcal{L}(D)^{-1}) - \chi^0(M, E \otimes \mathcal{L}(D)^{-1}) + \chi^0(M, E \otimes \mathcal{L}(D)^{-2})$$

Recursively continuing this, we get:

$$\chi^p(D, E) = \sum_{i=0}^{p} (-1)^i [\chi^{p-i}(M, E \otimes \mathcal{L}(D)^{-i}) - \chi^{p-i}(M, E \otimes \mathcal{L}(D)^{-(i+1)})]$$

This gives an important equation. These relation's do not necessarily hold for a general complex compact manifold, but if M is Kähler, or can be embedded into projective space, the relations hold, a key fact relying on GAGA results and the isomorphism between line bundles and bundles given by divisors as shown in [1].

Combining these together in a generating function, we get:

$$\chi_y(D, E) = \sum_{i=0}^{\infty} (-y)^i [\chi_y(M, E \otimes \mathcal{L}(D)^{-i}) - \chi_y(M, E \otimes \mathcal{L}(D)^{-(i+1)})]$$

2.1.4 Virtual χ_y-characteristic

skipping for now

2.2 Fundamental Kodaira Results

[3, p.138]

He now goes define a *Hodge Manifold* which is a Kähler manifolds whose fundamental class is in the image of the natural homomorphism $H^2(M, \mathbb{Z}) \to H^2(M, \mathbb{R})$. redundatnly, it was mentioned here that there are compact complex manifolds that are not Kähler. There are Kähler manifolds that are not Hodge, and but all algebraic manifold are Hodge. Note this definition is done by defining a certain order which I omitted.

Theorem 2.2.1: Bertini Theorem

Let F be a projectively induced line bundle over the algebraic manifold M. Then a non-singular divisor D induces F, namely $F = \mathcal{L}(D)$.

This is actually a special case of the general Bertini theorem that you find when googling "Bertini Theorem".

> ***Proof*** **:**
> will not prove

Theorem 2.2.2: Characterizing Varieties Geoemtrically

A compact complex manifold is a smooth complete variety (Hirzebruch: algebraic manifold) if and only if it is a Hodge Manifold.

> ***Proof*** **:**
> not proven. It is a GAGA like result that is interesting.

Hirzebruch now states that we know that for two choices of complex vector bundles E, E' over an algebraic manifold M, it can be that $\dim H^0(M,E) \neq H^0(M,E)$. But $\chi(M,E)$ does not depend on the continuous vector bundle W and depends only on the Chern classes of E. He now introducing some vanishing theorems:

Theorem 2.2.3: Kodaira Vanishing Theorem

Let F be a complex line bundle over a compact complex manifold M. If F^{-1} is positive, then the cohomology group $H^i(M,F)$ vanish for all $i \neq n$.

I did not define positivity of a bundle, he did it at the beginning of this section

> ***Proof*** **:**
> proved by Kodaira

Using Serre duality, we get

Theorem 2.2.4: Kodaira Vanishing Theorem - Euler Characteristic

let F, M be as before and take $F \otimes K^{-1}$ for the canonical class K. If it is positive, then the groups $H^i(M,F)$ vanish for all $i > 0$, so that:

$$\dim h^0(M,F) = \chi(M,F)$$

> ***Proof*** **:**
> again Kodaira

Theorem 2.2.5: Kodaira Vanishing Theorem II

Let F be a complex line bundle over a hole manifold M, and let F' be a positive line bundle over M. Then

$$H^i(M, F \otimes (F')^k)$$

vanishg for all $i > 0$ and k sufficiently large.

Proof **:**
Kodaira.

From it we get:

Theorem 2.2.6: Characterizing Line Bundles On Algebraic Manifold

Let M be an algebraic manifold, F an analytic line bundle over M. Then there are projectively induced line bundles A, B where $F = A \otimes B^{-1}$. Thus, we have:

$$F = \mathcal{L}(A) \otimes \mathcal{L}(B)^{-1}$$

Proof **:**
Hirzbruch actually proves this, see [3, p.141, thm 18.2.5].

Hirzebruch now states that he will make no distinction between a Hodge manifold and an algebraic manifold. In particular, if a compact complex manifold M admits a Hodge metric, then M is algebraic.

Theorem 2.2.7: Line Bundles Are Algebraic Over Projective Space

let L be a $\mathbb{C}$-fiber bundle over M with $\mathbb{C}\mathrm{P}^r$ as fibers and $\mathrm{PGL}_{r+1}(\mathbb{C})$ as the structure group. Then it is an algebraic manifold.

Proof **:**
proved by Kodaira

We shall need a generization of the above for because we'll need the case where F is the flag manifold $F(q) = \mathrm{GL}_q(\mathbb{C})/\Delta(q, \mathbb{C})$ and L is the associated $\mathrm{GL}_q(\mathbb{C})$-bundle ξ over M.

Theorem 2.2.8: Borel Vanishing Theorem

Let L be a complex fibre bundle over the algebraic manifold M with an algebraic manifold as fibre and a connected structure group. Assume that the first Betti number of F is zero. Then L is itself algebraic.

***Proof* :**
Borel, Hirzebruch gives a reference in [3, p. 141]

2.2.1 virtual χ_y-characteristic for algebraic manifolds

here

2.3 L-Genus and Todd Genus

Let B be a commutative ring, and $Z, \alpha_1, ..., \alpha_n$ be some independent intermediates (label the collection of alpha's α), all of degree 1. Let σ_j be symmetric functions in α. Let

$$q_j = \sigma_j(\alpha)$$

Define the ring $\mathcal{B} = B[[Z; \alpha_1, \alpha_2, ..., \alpha_n]]$. Take the ring $\mathcal{P} = B[[Z; q_1, ..., q_n]]$. Then $\mathcal{P}$ contains units called *one-units* that is:

$$1 = \sum_{j \geq 1} b_j^J \qquad b \in B$$

If $Q(z)$ is a one-unit, then

2.3.1 Todd Class Fixes Commutative Diagram

The diagram:

$$\begin{tikzcd} H^*(X;\mathbb{Q}) \arrow[r, "f_*"] & H^*(X;\mathbb{Q}) \\ K(X) \arrow[u, "\mathrm{ch}"] \arrow[r, "f_*"] & K(Y) \arrow[u, "\mathrm{ch}"'] \end{tikzcd}$$

Is not commutative, but the diagram:

$$\begin{tikzcd} H^*(X;\mathbb{Q}) \arrow[r, "f_*"] & H^*(X;\mathbb{Q}) \\ K(X) \arrow[u, "\mathrm{td}(X)\mathrm{ch}"] \arrow[r, "f_*"] & K(Y) \arrow[u, "\mathrm{td}(X)\mathrm{ch}"'] \end{tikzcd}$$

So the Chern character is in some way an insufficient tool to naturally represent homological information of vector bundles, but "fixing" it by the Todd class is.

2.3.2 Multiplicative sequences

As we saw with Chern and Pontryagin classes, their product translates to a cup product. This gives a natural algebraic relation that we know codify for future reference. Let B be a commutative ring, and define:

$$\mathcal{B} = B[z, p_1, p_2, ...]$$

Define a grading given by the following:

$$p_{j_1}p_{j_2}\cdots p_{j_r} \text{ has weight } j_1 + j_2 + \cdots + j_r$$

Then let $\mathcal{B}_k$ be the weight k-groups; note $\mathcal{B}_0 = B$ and $\mathcal{B}_r\mathcal{B}_s = \mathcal{B}_{r+s}$. Let $\pi(k)$ be all partitions of k. Then $\mathcal{B}_k$ is a B-module of rank $\pi(k)$.

Definition 2.3.1: Multiplicative Sequence

Let $\{K_j\}$ be a sequence of polynomials in indeterminates p_i where $K_0 = 1$ and $K_j \in \mathcal{B}_j$. Then the sequence $\{K_j\}$ is called a *multiplicative sequence* if whenver a power series is the product of two power series:

$$1 + p_1z + p_2z^2 + \cdots = (1 + p_1'z + p_2'z^2 + \cdots)(1 + p_1''z + p_2''z^2 + \cdots)$$

with indeterminates z, p_i', p_i'', then we get the identify:

$$\sum_{j=0}^{\infty} K_j(p_1, p_2, ...p_j)z^j = \left(\sum_{i=1}^{\infty} K_i(p_1', p_2', ...p_i')z^i\right)\left(\sum_{j=1}^{\infty} K_j(p_1'', p_2'', ...p_i'')z^j\right)$$

that is:

$$K\left(\sum_{i=0}^{\infty} q_iz^i\right) = \sum_{i=0}^{\infty} K_i(q_1, ..., q_i)z^i$$

where $q_i \in \mathcal{B}$ of weight i. In other words a sequence $\{K_j\}$ is called multiplicative if it induces an endomorphism on

$$B[p_1, p_2, ...]\,[[z]]$$

Given a multiplicative sequence $\{K_j\}$, we associate to it its *characteristic power series* as the image of $K(1+z)$:

$$K(1+z) = \sum_{j}^{\infty} K_j(1, 0, ..., 0)z^j = \sum_{j}^{\infty} b_jz^j$$

Recall we can define Chern roots. Hence we will be considering:

$$1 + p_1z + \cdots + p_nz^n = \prod_{i}^{n}(1 + \beta_iz)$$

where β_i is a symmetric polynomial in the p_i's. By some elementary ring theory and theory of symmetric functions, we can interpret $\mathcal{B}$ as the ring of all symmetric functions in $\beta_1, ..., \beta_n$.

We can characterize m-sequences via 1-unit power series:

Proposition 2.3.2: Correspondence 1-unit Power Series and m-sequences

There is a bijective correspondence between m-sequecnes and formal power series with constant term 1.

***Proof* :**
First, let's show any m-sequence $\{K_j\}$ is completely determined by its characteristic power series

$$Q(z) = K(1+z)$$

Proof. By decomposing:

$$1 + p_1 z + \cdots + p_n z^n = \prod_i^n (1 + \beta_i z)$$

We get:

$$\sum_{j=0}^{m} K_j(p_1, \ldots, p_j) z^j + \sum_{j=m+1}^{\infty} K_j(p_1, \ldots, p_m, 0, \ldots, 0) z^j = \prod_{i=1}^{m} Q(\beta_i z). \quad (6_m)$$

Therefore any polynomial K_j with $j \leq m$ is determined as a symmetric polynomial in the β_i and hence as a polynomial in the p_i. This holds for arbitrary m, completing the proof. □

Next, let

$$Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i \qquad b_i \in B$$

be a formal power series. Then there is an associated m-sequence $\{K_j\}$ where $K(1+z) = Q(z)$.

Proof. Consider the usual decomposition:

$$1 + p_1 z + \cdots + p_n z^n = \prod_i^n (1 + \beta_i z)$$

and consider the product $\prod_{i=1}^m Q(\beta_i z)$ where β_i are symmetric functions. Then the coefficients z^j are homogeneous of weight j. Thus, it is in $K_j^{(m)}(p_1, \ldots, p_j)$. It follows that $K_j^{(m)}$ does not depend on m for $m \geq j$.

Now, define $K_j = K_j^{(m)}$ for $m \geq j$. I claim $\{K_j\}$ is the required m-sequence. Indeed, the earlier proof the multiplicative property is true if the indeterminates p_i', p_i'' are replaced by 0 for large values of i, and so $\{K_j\}$ is an m-sequence. Finally applying $m = 1$ in equation:

$$\sum_{j=0}^{m} K_j(p_1, \ldots, p_j) z^j + \sum_{j=m+1}^{\infty} K_j(p_1, \ldots, p_m, 0, \ldots, 0) z^j = \prod_{i=1}^{m} Q(\beta_i z). \quad (6_m)$$

shows that $K(1+z) = Q(z)$, completing the proof. □

Combining these, we get the correspondence

The simplest example would be $\{p_j\}$ which has the characteristic series $1+z$. The notation p_i is due to the connection to Pontryagin classes. We shall be working with Chern classes so the following will be a useful substitution. Let $c_0 = p_0 = 1$, $z = x^2$, and γ_i be such that $\beta_i = \gamma_i^2$. then:

$$\sum_{i=1}^{\infty} p_i(-z)^i \left(\sum_{j=0}^{\infty} c_j(-x)^j \right) \left(\sum_{i=0}^{\infty} c_i x^i \right)$$

Naturally, if $\{K_j(p_1, ..., p_j)\}$ is an m-sequence with $Q(z)$ as the characteristic power series, then let $\{\widetilde{K}_j(c_1, ...c_j)\}$ be an m-sequence where $\widetilde{Q}(x) = Q(x^2)$. Then:

$$K_j(p_1, ..., p_j) = \widetilde{K}_{2j}(c_1, ..., c_{2j})$$
$$\text{and}$$
$$0 = \widetilde{K}_{2j+1}(c_1, ..., c_{2j+1})$$

Thus, the m-sequence in c_i with the characteristic power series $1 + x^2$ is given by

$$1, 0, p_1, 0, p_2, ...$$

From this relations, we get:

$$p_1 = -2c_2 + c_1^2$$
$$p_2 = 2c_4 - 2c_3c_1 + c_2^2$$
$$p_3 = -2c_6 + 2c + 5c + 1 = 2c_4c_2 + c_3^2, ...$$

Let's now find a way to calculate the explicit calculation of the polynomials of m-sequences. Given $Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i$, consider the factorization:

$$1 + b_1z + b_2z^2 + \cdots + b_mz^m = (1 + \beta_1' z) \cdots (1 + \beta_m' z)$$

Then define the sum:

$$\sum (\beta_1')^{j_1} (\beta_2')^{j_2} (\beta_r')^{j_r} \qquad \text{where} \qquad j_1 \geq j_2 \geq \cdots \geq j_r \geq 1, \ \sum_{s=1}^{r} j_s = k \leq m$$

which denotes the symmetric function in the β_i' which is the sum of all pairwise distinct monomials obtained by permuting $(\beta_1', ..., \beta_m')$ to the monomial $(\beta_1')^{j_1} (\beta_2')^{j_2} (\beta_r')^{j_r}$. The number of monomial in the sum is $m!/h$, h being the number of permutations living the aforementioned monomial fixed. Now:

Lemma 2.3.3: Computation of m-sequence

Let $\{K_j(p_1...., p_j)\}$ be the m-sequence corresponding to $Q(z) = 1 + \sum_{i=1}^{\infty} b_i z^i$. Then the coefficient of $p_{j_1}p_{j_2} \cdots p_{j_r}$ of K_s where $j_1 \geq j_2 \geq \cdots \geq j_r \geq 1$ and $\sum_{s=1}^{r} j_s = k$ is equal to:

$$\sum (j_1, j_2, ..., j_r)$$

Proof **:**
it is computations that will be omitted.

We shall want to define the m-sequence associated to Pontryagin classes associated to $\mathbb{C}\mathrm{P}^n$ seen as the classifying space for n-dimensional complex vector bundles. Recall that $\chi(\mathbb{C}\mathrm{P}^1 \oplus \mathbb{C}\mathrm{P}^2) = \chi(\mathbb{C}\mathrm{P}^1)\chi(\mathbb{C}\mathrm{P}^1)$. And that $\chi(\mathbb{C}\mathrm{P}^n) = 1$. We shall want to capture these properties using the Pontryagin class. Define:

$$Q(z) = \frac{\sqrt{z}}{\tanh\sqrt{z}} = 1 + \sum_{k=1}^{\infty}(-1)^{k-1}\frac{2^{2k}}{(2k)!}B_k z^k$$

where the B_k are the Bernoulli numbers. Note that in this case $B = \mathbb{Q}$. Furthermore, we shall have $B_k > 0$ skip the values of $1/2$. Thus we get:

$$B_1 = \frac{1}{6},\ B_2 = \frac{1}{30},\ B_3 = \frac{1}{42},\ B-4 = \frac{1}{30},\ B_5 = \frac{5}{66},\ B_6 = \frac{691}{2730},\ B_7 = \frac{7}{6},\ B_8 = \frac{3617}{510},...$$

We sahll denote the m-sequence of $Q(z)$ as $\{L_j(p_1, ..., p_j)\}$. Computing we get:

$$\begin{aligned}
L_1 &= \frac{1}{3}p_1 \\
L_2 &= \frac{1}{5\cdot 7}(7p_2 - p_1^2) \\
L_3 &= \frac{1}{3^3\cdot 5\cdot 7}(62p_3 - 13p_2p_1 + 2p_1^3) \\
&\vdots
\end{aligned}$$

We will want the following normalization: $L_k(p_1...., p_k) = 1$. To that end, if we let $p_i = \binom{2k+1}{i}$, then we get:

$$1 + p_1z + p_2z^2 + \cdots + p_kz^k = (1+z)^{2k+1} \bmod z^k + 1$$

From this we get:

Proposition 2.3.4: Coefficient of L-genus Power Series

Let $Q(z) = \frac{\sqrt{z}}{\tanh\sqrt{z}}$. Then for every k, the coefficient J_k of z^k in $(Q(z))^{2k+1}$ is equal to 1, and $Q(z)$ is the *only* power series with rational coefficient with this property.

Proof **:**
Apply Cauchy's integral formula and compute (see [3, p.12] to see the exact computations)

The above is useful for Pontryagin classes. We also requires the equivalent for Chern classes. Let:

$$Q(x) = \frac{x}{1-e^{-x}} = 1 + \frac{1}{2}x + \sum_{k=1}^{\infty}(-1)^{k-1}\frac{B_k}{(2k)!}x^{2k}$$

Then we get polynomials T_k called *Todd polynomials*. For future reference, we have the identity:

$$\frac{x}{1-\exp(-x)} = \exp\left(\frac{1}{2}x\right)\frac{\frac{1}{2}x}{\sinh\frac{1}{2}x}$$

By using the translation from Pontryagin to Chern class lemma, we get:

$$T+k(c_1,...,c_k) = \sum_{\substack{r,s \\ r+2s=k}} \frac{1}{2^{4s}r!}\left(\frac{1}{2}c_1\right)^r A_s(p_1,...,p_s)$$

We can compute the first few values:

$$\begin{aligned}
T_1 &= \frac{1}{2}c_1 \\
T_2 &= \frac{1}{12}(c_2+c_1^2) \\
T_3 &= \frac{1}{24}c_2c_1 \\
T_4 &= \frac{1}{720}(-c_4+c_3c_1+3c_2^2+4c_2c_1^2-c_1^4) \\
T_5 &= \frac{1}{1440}\left(-c_4c_1+c_3c_1^2+3c_2^2c_1-c_2c_1^3\right) \\
&\vdots
\end{aligned}$$

Again, by substituting $c_i = \binom{n+i}{i}$ defined by:

$$1+c_1x+\cdots+c_nx^n = (1+x)^{n+1} \bmod x^{n+1}$$

gives

$$T_n(c_1,...,c_n) = 1$$

Just like before, $Q(x)$ is the only power sires with rational coefficients with the property that $Q(x))^{k+1} = 1$ (see [3, p.14], or just repeat the above results with the appropriate modifications).

We shall need a special case. Let $B = \mathbb{Q}[y]$. Then consider the m-sequence $T_j(y;c_1,...,c_j)$ with characteristic power series:

$$Q(y;x) = \frac{x(y+1)}{1-e^{-x(y+1)}} - yx = \frac{x(y+1)}{e^{x(y+1)}-1} + x$$

Lemma 2.3.5

For every n the coefficient of x^n in $(Q(y;x))^{n+1}$ is equal to $\sum_{i=0}^{n}(-1)^i y^i$, and $Q(y,x)$ is the only power series with coefficients in $\mathbb{Q}[y]$ which has this property.

***Proof* :**
Take substitution

$$c_i = \binom{n+1}{i}$$

then:

$$T_n(y;c_1,\ldots,c_n) = 1-y+y^2-\cdots+(-1)^n y^n.$$

Then as $Q(\frac{1}{y}; yx) = Q(y; -x)$:

$$y^n T_n(\frac{1}{y}; c_1, \ldots, c_n) = (-1)^n T_n(y; c_1, \ldots, c_n)$$

as we sought to show

The polynomial $T_n(y; c_1, \ldots, c_n)$ can be written in a unique way in the form

$$T_n(y; c_1, \ldots, c_n) = \sum_{p=0}^{n} T_n^p(c_1, \ldots, c_n) y^p.$$

The polynomials $T_n^p(c_1, \ldots, c_n)$ satisfy

$$T_n^p(c_1, \ldots, c_n) = (-1)^n T_n^{n-p}(c_1, \ldots, c_n)$$

more on p.15

2.3.3 Todd Genus

Let X be a complex smooth manifold, and ξ a continuous q-vector bundle on X. Let $c_i \in H^{2i}(X, \mathbb{Z})$ be its Chern classes. Then the *total Todd class* of ξ is :

$$\mathrm{td}(\xi) = \sum_{j=1}^{\infty} T_j(c_1, ..., c_j)$$

where $\{T_j(c_1, ..., c_j)\}$ is given by the characteristic series:

$$Q(x) = \frac{x}{1 - e^{-x}} = 1 = \frac{1}{2}x + \sum_{j=1}^{\infty} (-1)^{j-1} \frac{B_j}{(2j)!} x^{2j}$$

where B_j is the Bernoulli numbers. To write down the m-sequence, recall that the m-sequence of the characteristic power series $Q(z) = \frac{2\sqrt{z}}{\sinh(2\sqrt{z}}$ is has as the first few terms:

$$A_1 = \frac{-2}{3},\ A_2 = \frac{2}{45}(-4p_2 + 7p_1^2)\ A_3 = \frac{-4}{3^2 \cdot 5 \cdot 7}\left(16p_3 - 44p_2p_1 + 31p_1^3\right)$$

Then note that :

$$\frac{x}{1 - e^{-x}} = \exp\left(\frac{1}{2}x\right) \frac{\frac{1}{2}x}{\sinh \frac{1}{2}x}$$

Then:

$$T_j(c_1, ..., c_j) = \sum \frac{1}{2^{4s} r!} \left(\frac{1}{2}c_1\right)^r A_s(p_1, ..., p_j)$$

where $r+2s=j$ and p_i are the Pontryagin classes. Recall this gives us:

$$\begin{aligned}
T_1(c_1) &= \frac{1}{2}c_1 \\
T_2(c_1,c_2) &= \frac{1}{12}(c_2+c_1^2) \\
T_3(c_1,c_2,c_3) &= \frac{1}{24}c_2c_1 \\
T_4(c_1,c_2,c_3,c_4) &= \frac{1}{720}(-c_4+c_3c_1+3c_2^2+4c_2c_1^2-c_1^4) \\
T_5(c_1,c_2,c_3,c_4,c_5) &= \frac{1}{1440}\left(-c_4c_1+c_3c_1^2+3c_2^2c_1-c_2c_1^3\right) \\
&\vdots
\end{aligned}$$

Then by our earlier work:

$$\mathrm{td}(\xi\oplus\xi') = \mathrm{td}(\xi)\mathrm{td}(\xi')$$

If $q=1$ and $c_1(\xi)=d\in H^2(X,\mathbb{Z})$, then:

$$\mathrm{td}(\xi) = \frac{d}{1-e^{-d}}$$

For future reference, we shall require the following factorization Given the factorization:

$$\sum_{j=0}^{q} c_j x^j = \prod_{i=1}^{q}(1+\gamma_i x)$$

Then we get:

$$\mathrm{td}(\xi) = \prod_{i=1}^{q}\frac{\gamma_i}{1-e^{-\gamma_i}}$$

Similarly, we can define the *total Chern Character* (not class):

$$\chi(\xi) = \sum_{i=1}^{q} e^{\gamma_i}$$

Then we get:

$$\begin{aligned}
\mathrm{ch}(\xi\oplus\xi') &= \mathrm{ch}(\xi)+\mathrm{ch}(\xi') \\
\mathrm{ch}(\xi\otimes\xi') &= \mathrm{ch}(\xi)\mathrm{ch}(\xi')
\end{aligned}$$

If $q=1$ and $c_1(\xi)=d\in H^2(X,\mathbb{Z})$, then we get $\mathrm{ch}(\xi)=e^d$. In general, $\chi(\xi)=q+\sum_{k=1}^{\infty}\mathrm{ch}_k(\xi)$ where

$$\chi_k(\xi) = \frac{s_k}{k!}H^{2k}(x,\mathbb{Z})\otimes\mathbb{Q} \qquad s_k = \sum_{i=1}^{q}\gamma_i^k$$

Theorem 2.3.6: Relating Todd Class and Chern Character

Let ξ be a continuous $\mathrm{GL}_q(\mathbb{C})$-bundle over X. Then:

$$\sum_{r=0}^{q}(-1)^r \mathrm{ch}\left(\bigwedge^r \xi^*\right) = (\mathrm{td}(\xi))^{-1}c_q(\xi)$$

***Proof* :**
Proof: If $\sum_{j=0}^q c_j(\xi)x^j = \prod_{i=1}^q(1+\gamma_i x)$ by the properties of Chern classes (though, I did not actually prove this result in my vect. bundle notes. thm 4.4.3 in Hirzebruch, p.64)

$$\mathrm{ch}\bigwedge^r \xi^* = \sum e^{-(\gamma_{i_1}+\cdots+\gamma_{i_r})}$$

where the sum is over all combinations $i_1, \ldots, i_r$ with $1 \leq i_1 < \cdots < i_r \leq q$. Therefore

$$\sum_{r=0}^{q}(-1)^r \mathrm{ch}\bigwedge^r \xi^* = \prod_{i=1}^{q}(1-e^{-\gamma_i}) = (\gamma_1\cdots\gamma_q)\prod_{i=1}^{q}\frac{1-e^{-\gamma_i}}{\gamma_i} = (\mathrm{td}(\xi))^{-1}c_q(\xi).$$

Now, recall that if we are talking about the Chern class of a complex manifold M, we are talking about the Chern class of its tangent bundle, and the K-genus is defined for such structures. Let

$$Q(x) = \frac{x}{1-e^{-x}} \qquad \text{and} \qquad Q(y;x) = \frac{x(y+1)}{1-e^{-x(y+1)}} - xy$$

Then:

Definition 2.3.7: Todd Genus

Let M_n be an n-dimensional almost complex manifold. The *Todd genus* (or *T-genus*) is the number:

$$T_n(M_n) = T_n(c_1, ..., c_n)[M_n]$$

As usual, the $[M_n]$ represents the tangent bundle of M_n. The reader can verify that the direct product of manifolds $X \times Y$ is multiplicative on the Todd class. This means that if

$$0 \to \xi' \to \xi \to \xi'' \to 0$$

Then:

$$(1+c_1't+\cdots+c_{q'}'t^{q'})(1+c_1''t+\cdots+c_{q''}''t^{q''}) = 1+c_1t+\cdots+c_qt^q$$

and so:

$$\mathrm{td}(\xi')(t)\cdot\mathrm{td}(\xi'')(t) = \mathrm{td}(\xi)(t)$$

In many ways, the Todd class is "dual" to the Chern class. For example ([3, p.93]):

$$T_{-1}(M)n) = \sum_{p=0}^{n}(-1)^p T^p(M_n) = c_n[M_n]$$

and:

$$T_1(M)n) = \sum_{p=0}^{n} T^p(M_n) = \tau(M_n)$$

where τ is the index of M_n. More generally:

Proposition 2.3.8: Properties of Todd Genus

Let B be a commutative ring where $Q \subseteq B$. Then

1. $\sum_{l=0}^{n} T_n^{(l)}(c_1, \ldots, c_n) = c_n$, for all n.
2. $T_n^{(0)}(c_1, \ldots, c_n) = \mathrm{td}(c_1, \ldots, c_n)(= T_n(c_1, \ldots, c_n))$.
3. $$\sum_{l=0}^{n} T_n^{(l)}(c_1, \ldots, c_n) = \begin{cases} 0 & \text{if } n \text{ is odd} \\ L_{\frac{n}{2}}(p_1, \ldots, p_{\frac{n}{2}}) & \text{if } n \text{ is even.} \end{cases}$$

The *total Todd class* of a vector bundle, ξ, is

$$\mathrm{td}(\xi)(t) = \sum_{j=0}^{\infty} \mathrm{td}_j(c_1, \ldots, c_j)t^j = 1 + \frac{1}{2}c_1(\xi) + \frac{1}{12}(c_1^2(\xi) + c_2(\xi))t^2 + \frac{1}{24}(c_1(\xi)c_2(\xi))t^3 + \cdots$$

Proof **:**
exercise

Now, the Chern character is a bit different. If :

$$0 \to \xi'' \to \xi \to \xi' \to 0$$

then:

$$\mathrm{ch}(\xi)(t) = \mathrm{ch}(\xi')(t) + \mathrm{ch}(\xi'')(t)$$

And similarly Here's the LaTeX code for the equation:

$$\mathrm{ch}(\xi \otimes \eta)(t) = \sum_{i,j} e^{(\gamma_j + \delta_j)t} = \sum_{i,j} e^{\gamma_j t} e^{\delta_j t} = \left(\sum_i e^{\gamma_j t}\right)\left(\sum_j e^{\gamma_j t}\right) = \mathrm{ch}(\xi)(t)\,\mathrm{ch}(\eta)(t).$$

Hence, it defines a ring homomorphism $K(X) \to H^*(X, \mathbb{Q})$.

Definition 2.3.9: T-characteristic

Let X be an admissible complex manifold, ξ a $U(q)$-vector bundle. Then

$$T(X, \xi)(t) = \chi(\xi)(t)\mathrm{td}(\xi)(t)$$

Is called the *T-characteristic* of ξ over X (see [4, p.226]).

The T-characteristic satisfies the duality:

$$(-1)^n T_n^{(l)}(c_1, ..., c_n) T_n^{(n-l)}(c_1, ..., c_n)$$

2.3.4 Pontryagin and Chern Numbers

Let V^n be an oriented compact differentiable manifold where n is the dimension. As V is compact and orientable, it is a cycle, and by cohomology theory V^n is generated by one element. The value of the n-dimensional cohomology class x on the fundamental cycle V^n is denoted $x[V^n]$. if the coefficeints of the cohomology group are in A, then:

$$x \in H^n(V^n, A) \qquad x[V^n] \in A$$

We can extend this by tensoring by a group B so that :

$$x[V^n] \in A \otimes B \qquad x \in H^n(V^n, A) \otimes B$$

For our purposes, let $n = 4k$ and let $p_i \in H^{4i}(V^{4k}, \mathbb{Z})$ be the Pontryagin classes of V^{4k} (i.e. of its tangent bundle). We get a natural grading by the indices i so that given $k = j_1 + j_2 + \cdots + j_r$ we get the product $p_{j_1} \cdots p_{j_r}$ and an integer (as $A = \mathbb{Z}$):

$$p_{j_1} \cdots p_{j_r}[V^{4k}]$$

Let $\pi(k)$ be the number of distinct partitions of k.

Definition 2.3.10: Pontryagin Numbers and Chern Number

The *Pontryagin numbers* are the $\pi(k)$ numbers :

$$p_{j_1} \cdots p_{j_r}[V^{4k}]$$

where $n = 4k$. If n is not divisible by 4, then the Pontryagin number is 0.

Similarly the *Chern numbers* of a complex manifold M (with an almost complex structure so that we are given a particular orientation) are all:

$$c_{j_1} \cdots c_{j_r}[V^{2k}]$$

where $n = j_1 + j_2 + \cdots + j_r$ and the j_i run through all partitions.

By some characteristic class theory, if M_n is an oriented almost complex manifold, $c_n[M_n] = \chi(M_n, TM_n)$. We shall focus on Pontryagin numbers as we shall need them to prove an important result in cohomology, and return to Chern numbers once we cover the Todd genus in section 2.3.3.

We can define an equivalence relation where two manifolds $M \sim_p N$ if they have the same dimension and are the same Pontryagin number. Note that if the dimension is not divisible by 4, all manifolds are in the same equivalence relation given by $\sim_p$ as they are all 0.

Now let M, N be smooth $n = 4k$ dimensional manifolds. Then by classical differential geometry we have $T(M \times N) = \pi_M^*(TM) \times \pi_N^*(TM)$ (consider for now the real tangent bundle, so it is a

$\mathrm{GL}_n(R)$-bundle). Then considering the Pontryagin classes of M, N and $M \times N$, by the properties of characteristic classes we get:

$$1 + p_1'' + p_2'' + \cdots = (1 + p_1 + p_2 + \cdots)(1 + p_1' + p_2' + \cdots)$$

Or, as a generating function:

$$\sum_{k=0}^{\infty} p_k'' z^k = \sum_{i=0}^{\infty} f^*(p_i) z^i \sum_{j=0}^{\infty} g^*(p_j') z^j \text{ mod torsion.}$$

In terms of the fundamental class:

$$(f^*(x)g^*(y))[M \times N] = x[M] \cdot y[N] \qquad \forall x \in H^n(M, \mathbb{Z}) \otimes B,\ y \in H^m(N, \mathbb{Z}) \otimes B$$

This last property is simply the properties of m-sequences:

Lemma 2.3.11: Pontryagin Character and M-sequence

HERE

Proof **:**

Definition 2.3.12: K-genus

Let V^n be an oriented n-dimensional manifolds and $\{K_j(p_1, ..., p_j)\}$ be an m-sequence where $K(V^{4k}) = K_k[V^{4k}]$ and 0 otherwise. Then:

$$\begin{cases} 0 & \dim_{\mathbb{R}} X \not\equiv 0 \text{ mod } 4 \\ K_n(p_1, ..., p_n)[X] & \dim_{\mathbb{R}} X = 4n \end{cases}$$

is called the *K-genus* of V^n.

If we are working with Chern classes, then we would have $n = 2k$.

By what we've discussed earlier, the K-genus is multiplicative.

Example 2.3.13: K-genus

look at [3, p.77].

1. L-genus defined earlier
2. A-genus

2.4 Some Cobordism

Pontryagin classes will be important characterizing cobordism, and we shall need a bit of cobordism in the proof of HRR. What we shall show that that cobordant manifolds are characterized by their Pontryagin class. We start by creating the algebraic structure that are naturally induced by Pontryagin classes and then quote the result by Thom that shows it is isomorphic to the cobordant ring.

Definition 2.4.1: Algebraic Cobordism Ring

Let V^n, W^n be two oriented manifolds of the same dimension. Let $-V^n$ represent the reverse orientation. It is easy to see that:

$$K(V^n \sqcup W^n) = K(V^n) + K(W^n) \qquad K(-V^n) = -K(V^n)$$

Thus, $K(-)$ is compatible with the equivalence relation $\sim_p$ and creates an additive group $\widetilde{\Omega}^n$. Note that $\widetilde{\Omega}^n = 0$ if $4 \nmid n$. Let $\widetilde{\Omega} \bigoplus_n \widetilde{\Omega}^n$ so that by the above equations we have the natural grading:

$$\widetilde{\Omega}^n \widetilde{\Omega}^m \subseteq \widetilde{\Omega}^{nm}$$

And so $\widetilde{\Omega}$ is a commutative graded (torsion-free) ring.

The ring was called "algebraic" as it was defined using the algebraic properties of Pontryagin classes, we shall soon define the "geometric" cobordant ring that is the equivalence classes of cobordant manifolds and show they are isomorphic. For now, let us characterize the algebraic structure of the algebraic cobordism ring: the goal is to show that

$$\widetilde{\Omega} \otimes \mathbb{Q} \cong \mathbb{Q}[x_1, x_2, ...]$$

To that end, we must find which elements of $\widetilde{\Omega}$ can represent each x_i. There is no natural choice, hence we define "candidate sequences" of elements given by manifolds $V^4, V^8, V^{12}, ...$ that can be chosen to form this basis. We shall show these can be given by complex projective spaces of even dimension. Let us start by recalling that the Pontryagin class of V^{4k} can be written in indeterminate z and decomposed into:

$$1 + p_1 z + \cdots + p_k z^k = \prod_{i=1}^{k} (1 + \beta_i z)$$

Define:

$$s(V^k) = (\beta_1^k + \cdots + \beta_k^k)[V^{4k}]$$

By some symmetry function theory, we can think of this as needing the appropriate homogeneity and dimension properties We shall define a *basis sequence* a sequence where:

$$s(V^{4k}) \neq 0$$

Theorem 2.4.2: Basis Sequence Correspondence

Let $\{V^{4k}\}$ be a basis sequence and B a [commutative] ring containing $\mathbb{Q}$. Then the set of sequences $\{a_k\}_k$ where $a_k \in B$ are in bijective correspondence to the set of m-sequences $\{K_j(p_1, ..., p_j)\}$ with coefficients in B where $K(V^{4k}) = a_k$.

***Proof* :**
[3, p.79] (but it is simple)

Theorem 2.4.3: Complex Projective Forms Basis Sequence

Let $\mathbb{C}\mathrm{P}^{2k}$ be the $2k$-dimensional complex projective space. Then they form a basis sequence where

$$s(\mathbb{C}\mathrm{P}^{2k+1}) = 2k+1$$

***Proof* :**
ibid; simple proof.

Let us now consider $\widetilde{\Omega}\otimes\mathbb{Q}$, so now $[V^{4k}]$ can be written as $\frac{a}{b}[V^{4k}]$. Certainly the Pontryagin number, K-genus, and $s(V^{4k})$ are all still well-defined (where the K-genus is well-defined as we required B to contain $\mathbb{Q}$). Now, the Pontryagin number of an element of $\widetilde{\Omega}\otimes\mathbb{Q}$ can be a non-integer. This new ring can still be represented using a basis sequence:

Theorem 2.4.4: Basis Sequence With Rational Numbers Correspondence

Let $\{V^{4k}\}$ be a basis sequence of oriented manifolds, and let $(j) = (j_1, ..., j_r)$ be a partition of k. Let $V_{(j)} = V^{4j_1} \times \cdots \times V^{4j_r}$. Then every $\alpha \in \widetilde{\Omega}\otimes\mathbb{Q}$ can be uniquely represented by the sum:

$$\alpha = \sum_{(j)\in\pi(j)} r_{(j)}[V_{(j)}] \qquad r_{(j)} \in \mathbb{Q}$$

where the sum goes over all partitions of k. Furthermore, given a system $\{a_{(j)}\}$ of rational numbers, there exists an element $\alpha \in \widetilde{\Omega}\otimes\mathbb{Q}$ where

$$p_{(j)}[\alpha] = \alpha_{(j)}$$

***Proof* :**
[3, p.80] (This is some linear algebra (recall the proof in comm. alg. using the Vandermonde Determinant)

Theorem 2.4.5: Isomorphism of Algebraic Cobordism Ring

$$\widetilde{\Omega}\otimes\mathbb{Q} \cong \mathbb{Q}[x_1, x_2, ...]$$

***Proof* :**
Let $\{V^{4k}\}$ be an arbitrary sequence of oriented manifolds. What's left to show is that:

1. given $\alpha = \sum_{(j)} r_{(j)}[V_{(j)}]$ then $s(\alpha) = r_k s[V^{4k}]$
2. If for all k, every element of $\alpha \in \widetilde{\Omega}^{4k}\otimes\mathbb{Q}$ can be represented as a sum $\alpha = \sum_{(j)}[V_{(j)}]$,

then $\{V^{4k}\}$ is a basis sequence

[3, p.80]

There is a natural question of smallest integers can represent a system of Pontryagin numbers and it invovles the μ that I skipped.

Lastly, the K-genus can be characterized as being any homomorphism from the algebraic cobordism ring to $\mathbb{Q}$:

Theorem 2.4.6: Characterizing Rational Homomorphisms

highlight link @markup.link.label texRefArg Let $\varphi : \widetilde{\Omega} \otimes \mathbb{Q} \to \mathbb{Q}$ be a homomorphism. Then there exists a K such that $\varphi(-) = K(-)$.

Proof **:**
easy on p.81

Definition 2.4.7: Geometric Cobordism Ring

Two oriented smooth n-dimensional manifolds M^n, N^n are *cobordant* if there exist a manifold C such that
$$\partial C \cong (M^n \sqcup -N^n)$$
where $-N^n$ is the reverse orientation; if such a C exists it defines an equivalence relation $M \sim_c N$ which respects disjoint union, reverse orientation, and direct product; $\sqcup, -, \times$. Then Ω^n is the collection of such equivalence classes and:
$$\Omega = \bigoplus_n \Omega^n$$
is the *geometric cobordism ring*.

Certainly this ring is graded, and it is *anti-commutative*, namely:
$$\alpha \cdot \beta = (-1)\beta \cdot \alpha \qquad \alpha \in \Omega^n,\ \beta \in \Omega^m$$

He now goes onto to prove the isomorphism between this ring and the algebraic one, skipping for now.

An important theorem:

Theorem 2.4.8: Hirzebruch Signature Theorem

Let M be a real differentiable oriented manifold, and let $I(M)$ be its index. Then:

1. if $\dim_{\mathbb{R}} M \not\equiv 0 \bmod 4$, $I(M) = 0$
2. if $\dim_{\mathbb{R}} M \equiv 0 \bmod 4$, then:

$$I(M) = L_k(p_1, ..., p_k)[M]$$

***Proof* :**
[4, p.231].

2.4.1 Index and Virtual index

[3, p.84-90]

2.5 RR for algebraic manifolds (i.e. projective verities)

Let M be an algebraic manifold, $T(M)$ be the Todd genus, and $\chi(M)$ be the arithmetic genus. We shall show that $T(V) = \chi(V)$. This then RR for complex line bundles over M.

Theorem 2.5.1: Splitting of Algebraic Manifold

Let M be an algebraic manifold which is also a complex split manifold. Then:

$$\chi(M) = T(M)$$

***Proof* :**
Let $m = \dim M$, and $A_1, ..., A_m$ the complex diagonal line bundles defined over M. Then by section 13.6 eq.(13):

$$(1+y)^m T(M) = \sum_{l=0}^{m} y^l \sum_{1 \leq i_1 < \cdots < i_l \leq n} T_y(a_{i_1}, ...a_{i_l})_M$$

where the T_y are polynomials, and by theorem 17.4.1 (in virtual char:

$$(1+y)^m \chi(V) = \sum_{l=0}^{m} y^l \sum_{i_1 < \cdots < i_l} \chi_V(A_{i_1}, \ldots, A_{i_l})_V$$

As M is algebraic, theorem 19.2.1 (vert char for alg man) χ_y is a polynomial. Then we see by the equation that the two agree if for some $y \neq -1$ (or else $1 + y = 0$) the two match. By choosing $y = 1$, theorem 19.5.1, literally last result of the section say they agree at $y = 1$, completing the proof.

Now, as we know we can pullback any vector bundle E over M to via the map $F(M) \to M$ where $F(M)$ is the flag manifold, by theorem 14.3.1 (saying the T-char equals on pullback we have $T(M) = T(F(M))$ (Hirzebruch denotes $F(M)$ as M^{Δ}). We just now need the corresponding result for the arithmetic genus:

Theorem 2.5.2: Euler-characteristic Invariant Under Splitting

Let ξ be a complex $\mathrm{GL}_q(\mathbb{C})$-bundle over an algebraic manifold M, $M' = F(M)$ the flag manifold. Let φ be the map and $\varphi^*(\xi)$ be the pullback. Then

$$\chi(M) = \chi(M')$$

Now I'm a bit confused, why no ξ, $\varphi^*(\xi)$ with the Eul.Char???

Proof **:**
[3, p.149], quote:

1. theorem 18.3.1*
2. theorem 15.9.1

Theorem 2.5.3: Todd and Arithmetic Genus Equivalence

Let M be an algebraic manifold. Then:

$$\chi(M) = T(M)$$

Proof **:**
string the above together!

The above can be thought as proving that $\chi_0(X) = T_0(M)$. Then theorem 19.5 implies:

Theorem 2.5.4: Extension To Line Splitting

Let M be an algebraic manifold, $F_1, ..., F_r$ complex line bundles over M. Then:

$$\chi(F_1, ..., F_r)_M = T(F_1, ..., F_r)_V$$

Proof **:**
This is quoting theorem 19.5

Note for $r = 1$, $F_1 = F$, $\chi(F)_M = T(F)_V$. Then combining theorem 17.2 and theorem 12.1(4):

$$\chi(F) = \chi(M) - \chi(M, F^{-1})$$

and:

$$T(F) = T(M) - T(M, F^{-1})$$

replacing F by F^{-1} theorem 20.2.2, theorem 20.3.1 implies:

$$\chi(M,F) = T(M,F)$$

But this *is* HRR for line bundles!! I will box this to make it obvious.

3

Organization Ideas

1. algebraic setting: quote important chern class theory
2. necessary cobordism background
3. geometric setting: Hodge theory, Kähler manifolds, some GAGA results and Hodge manifold, euler-characteristic and generalizations.

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